

unit-3



**Ganpat
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Department of
Computer Science



B.Sc. Hons. (CA&IT) Sem-II
FINANCIAL ACCOUNTING
UNIT-3
Accounting Ratios

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1. Accounting Ratio

- Accounting ratio shows the **relationship between two or more group of items in Balance sheet or profit & loss account statement.**
- They are used to analyze the financial performance and position of a business.

2. There are two classification of Accounting Ratio

Traditional Classification	Functional Classification
1) Revenue Statement Ratio Based on items from the Profit & Loss Account. (Example: Gross Profit Ratio) -Shows how much profit is earned after selling goods. 2) Balance Sheet Ratio Compares items from the Balance Sheet only. (Example: Current Ratio) -Shows whether the company can pay its short-term debts. 3) Composite Ratio Compares one item from P&L Account and one from Balance Sheet. (Example: Return on Investment) -Shows how well the money invested is being used.	1) Liquidity Ratios Show a firm's ability to pay short-term debts. Examples: Current Ratio, Liquid Ratio -Can the business pay its immediate bills? 2) Profitability Ratios Show how much profit the business is making. Examples: Gross Profit Ratio, Net Profit Ratio -Are we making enough money? 3) Leverage Ratios Show the proportion of owners' capital and borrowed capital. Examples: Debt-Equity Ratio, Proprietor's Ratio -How much loan is used compared to owner's money? 4) Activity/Efficiency Ratios Show how efficiently the business uses its assets.

	Examples: Stock Turnover Ratio, Debtors Turnover Ratio -Are we using our assets effectively?
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3. Advantages of Accounting Ratio

1) Profitability

- Shows how much profit the business is earning.
- Ratios like Gross Profit Ratio and Net Profit Ratio help measure profit levels.

2) Liquidity

- Ratios like Current Ratio and Liquid Ratio show if the business can pay its short-term debts.

3) Efficiency

- Turnover ratios show how efficiently the business uses its resources.
- Example: Stock Turnover Ratio shows how fast stock is sold.

4) Inter-Firm Comparison

- Helps compare ratios with other firms.
- Shows strengths and weaknesses of the business.

5) Trend Indication

- Ratios of the last 3–5 years show the performance trend of a business.

6) Useful for Budgetary Control

- Budgets include ratios to understand financial position and make corrections.

7) Useful for Decision-Making

- Management uses ratios to take business decisions.
- Example: If Liquid Ratio is low, the company may arrange more funds.

Limitations

Ratios are helpful, but they aren't perfect:

1. **Historical Data:** Ratios use past numbers. They don't always predict the future perfectly.
2. **Price Level Changes:** Inflation can make ratios misleading (e.g., assets bought 10 years ago look "cheap" on paper).

3. **Qualitative Factors:** A ratio can't measure employee morale, brand reputation, or management quality.

1) Traditional Classification :-

(1) Revenue statement Ratio's :-

- (i) Gross profit Ratio (GP Ratio)
- (ii) Net profit Ratio (NP Ratio)
- (iii) Operating Ratio
- (iv) Stock Turn over Ratio

(2) Balance Sheet Ratio :-

- (i) Current Ratio
- (ii) liquid Ratio
- (iii) Acid-test Ratio
- (iv) Debt - Equity Ratio

(3) composite Ratio :-

4. debtors Ratio (debtor turn over)
5. Creditors Ratio (creditor turn over)

6. Important Accounting Ratios

Ratio	Formula	Meaning
Revenue statement Ratio's :- To see if the business is generating a sufficient return on its efforts.		
Gross Profit Ratio (GPR)	$\frac{\text{Gross Profit} \times 100}{\text{Net Sales}}$	Shows profit from trading. It shows how much profit remains after deducting direct production costs (COGS).
Net Profit Ratio (NPR)	$\frac{\text{Net Profit} \times 100}{\text{Net Sales}}$	Shows overall profitability. The overall profitability after <i>all</i> expenses

		(taxes, interest, admin) are paid.
Operating Ratio	$\frac{\text{Cost of goods sold} + \text{Operating expenses}}{\text{Net sales}} \times 100$ Cost of goods sold = opening stock + net purchase + purchase expenses – closing stock Operating expenses = administrative expenses + interest + selling expenses	Shows efficiency of operations. focuses on what is spent to generate revenue
Stock Turnover Ratio	(Cost of Goods Sold ÷ Average Stock)	Shows efficiency of stock usage. How many times a year the company sells and replaces its inventory. A high ratio means products are selling fast.
Balance Sheet Ratio		
Current Ratio/Liquid Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$ Current assets = Cash & Bank balance + Stock + debtors + prepared expenses + loan & advanced + investment Convertible into cash. Current liability = Creditors + bank o/d + dividend + taxation + Unclaimed dividend	Short-term liquidity. For every \$1 rupee of debt due soon, how many rupees of assets do we have? A standard ratio is 2:1.
Liquid Ratio	$\frac{\text{Liquid Assets}}{\text{Liquid Liabilities}}$	Liquidity without stock
Acid Test Ratio	$\frac{\text{Quick Assets}}{\text{Liquid Liabilities}}$	While the Current Ratio looks at all current

	Quick Assets = Current Assets – (Stock + Prepaid Expenses) Quick Liabilities = Current Liabilities – Bank Overdraft (optional, depending on the firm's accounting policy)	assets, the Acid-Test Ratio focuses only on Quick Assets. These are assets that can be converted into cash almost immediately (usually within 90 days) without a significant loss in value. Crucially, it excludes Stock (Inventory) because selling stock takes time and is not guaranteed.
Debt Equity Ratio	$\frac{\text{Long term liabilities}}{\text{Share holders fund}} \times 100$	he balance between "Borrowed Money" (Debt) and "Owner's Money" (Equity). High debt increases financial risk.
composite Ratio		
Debtors Ratio / Debtors Turnover	$\frac{\text{Debtors} + \text{Bills Receivable}}{\text{Average daily Sales}}$ Average daily sales = $\frac{\text{Credit Sales}}{365}$ OR $\text{Debtors Ratio} = \frac{\text{Debtors} + \text{Bills Receivable} \times 365}{\text{Credit Sales}}$	Shows collection efficiency

Creditors Ratio / Creditors Turnover	Creditors Ratio = $\frac{(\text{Creditors} + \text{Bills Payable})}{\text{Average Daily Purchases}}$ Where: Average Daily Purchases = $\frac{\text{Credit Purchases}}{365}$ OR Creditors Ratio = $\frac{(\text{Creditors} + \text{Bills Payable})}{\text{Credit Purchases}} \times 365$	Shows payment period
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Cost of Goods Sold (COGS)	Sales - Gross Profit Cost of goods sold = opening stock + net purchase + purchase expenses - closing stock	Direct cost of producing goods sold by a business.
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❖ Cost of Goods Sold (COGS)

The direct cost of the products sold. If you sell a shirt for \$500\$, the COGS is the cost of the fabric and the tailor's wage (e.g., \$200\$). It does **not** include the shop's rent or electricity.

- Cost of Goods Sold means the **direct cost of producing goods sold by a business.**
- It includes **only direct costs, not indirect costs.**
- Direct costs change with the level of production.
- Examples of direct costs are:
 - Direct materials
 - Direct labour
- **Indirect and fixed costs like office expenses, administrative expenses, and marketing expenses are not included in COGS.**
- COGS may differ from one business or industry to another.

Cost of goods sold = opening stock + net purchase + purchase expenses - closing stock

Formula of Cost of Goods Sold

COGS = Sales - Gross Profit

Example

- Sales = ₹3,50,000
 - Gross Profit = ₹2,00,000
- COGS = 3,50,000 - 2,00,000 = ₹1,50,000
- 👉 Therefore, Cost of Goods Sold = ₹1,50,000

EXAMPLES

ACCOUNTING RATIOS - EXAMPLES FROM MATERIAL

PROBLEM 1

- Opening Stock: 70,000
- Closing Stock: 80,000
- Total Sales: 5,00,000
- Cost of Goods Sold (COGS): 3,00,000
- Gross Profit: 2,00,000
- Net Profit: 80,000

Required: Calculate the following ratios: (a) Gross Profit Ratio (b) Net Profit Ratio (c) Stock Turnover Ratio

--- SOLUTION ---

(a) Gross Profit Ratio

- **Formula:** $(\text{Gross Profit} / \text{Net Sales}) \times 100$
- **Calculation:** $= (2,00,000 / 5,00,000) \times 100 = 0.40 \times 100 = \mathbf{40\%}$

(b) Net Profit Ratio

- **Formula:** $(\text{Net Profit} / \text{Net Sales}) \times 100$
- **Calculation:** $= (80,000 / 5,00,000) \times 100 = 0.16 \times 100 = \mathbf{16\%}$

(c) Stock Turnover Ratio

- **Formula:** $\text{Cost of Goods Sold} / \text{Average Stock}$
 - **Step 1: Calculate Average Stock** $= (\text{Opening Stock} + \text{Closing Stock}) / 2 = (70,000 + 80,000) / 2 = 1,50,000 / 2 = 75,000$
 - **Step 2: Calculate Ratio** $= 3,00,000 / 75,000 = \mathbf{4 \text{ Times}}$
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PROBLEM 2

- Opening Stock: 50,000
- Closing Stock: 30,000
- Total Sales: 2,50,000
- Gross Profit: 80,000
- Net Profit: 50,000

Required: Calculate the following ratios: (a) Gross Profit Ratio (b) Net Profit Ratio (c) Stock Turnover Ratio

--- SOLUTION ---

(a) Gross Profit Ratio

- **Formula:** $(\text{Gross Profit} / \text{Net Sales}) \times 100$
- **Calculation:** $= (80,000 / 2,50,000) \times 100 = 0.32 \times 100 = \mathbf{32\%}$

(b) Net Profit Ratio

- **Formula:** $(\text{Net Profit} / \text{Net Sales}) \times 100$
- **Calculation:** $= (50,000 / 2,50,000) \times 100 = 0.20 \times 100 = \mathbf{20\%}$

(c) Stock Turnover Ratio

- **Formula:** $\text{Cost of Goods Sold} / \text{Average Stock}$
- **Step 1: Calculate Cost of Goods Sold (COGS)** Formula: $\text{Sales} - \text{Gross Profit} = 2,50,000 - 80,000 = 1,70,000$
- **Step 2: Calculate Average Stock** Formula: $(\text{Opening Stock} + \text{Closing Stock}) / 2 = (50,000 + 30,000) / 2 = 80,000 / 2 = 40,000$
- **Step 3: Calculate Ratio** $= 1,70,000 / 40,000 = \mathbf{4.25 \text{ Times}}$

QUESTION 3

From the following information, calculate:

1. Gross Profit Ratio
2. Net Profit Ratio
3. Operating Ratio

Information given:

- Net Sales = ₹5,00,000
- Gross Profit = ₹2,00,000
- Operating Expenses = ₹1,20,000

Calculate Cost of Goods Sold (COGS)

$$\begin{aligned}\text{COGS} &= \text{Sales} - \text{Gross Profit} \\ &= 5,00,000 - 2,00,000 = ₹3,00,000\end{aligned}$$

Gross Profit Ratio

$$\begin{aligned}\text{Gross Profit Ratio} &= \frac{\text{Gross Profit}}{\text{Net Sales}} \times 100 \\ &= \frac{2,00,000}{5,00,000} \times 100 = \boxed{40\%}\end{aligned}$$

Net Profit Ratio

: calculate Net Profit:

$$\begin{aligned}\text{Net Profit} &= \text{Gross Profit} - \text{Operating Expenses} \\ &= 2,00,000 - 1,20,000 = ₹80,000\end{aligned}$$

$$\text{Net Profit Ratio} = \frac{80,000}{5,00,000} \times 100 = \boxed{16\%}$$

Operating Ratio

$$\begin{aligned}\text{Operating Ratio} &= \frac{\text{COGS} + \text{Operating Expenses}}{\text{Net Sales}} \times 100 \\ &= \frac{3,00,000 + 1,20,000}{5,00,000} \times 100 \\ &= \frac{4,20,000}{5,00,000} \times 100 = \boxed{84\%}\end{aligned}$$

QUESTION 4

From the following Information, calculate:

1. Current Ratio
2. Liquid Ratio (Quick Ratio)

Information given:

- Cash = ₹15,000
- Stock = ₹25,000
- Debtors = ₹30,000
- Prepaid Expenses = ₹5,000
- Creditors = ₹40,000
- Outstanding Expenses = ₹10,000

Step 1: Calculate Current Assets

$$\text{Current Assets} = \text{Cash} + \text{Stock} + \text{Debtors} + \text{Prepaid Expenses}$$

$$= 15,000 + 25,000 + 30,000 + 5,000 = ₹75,000$$

Step 2: Calculate Current Liabilities

$$\text{Current Liabilities} = \text{Creditors} + \text{Outstanding Expenses}$$

$$= 40,000 + 10,000 = ₹50,000$$

CURRENT RATIO

$$\begin{aligned}\text{Current Ratio} &= \frac{\text{Current Assets}}{\text{Current Liabilities}} \\ &= \frac{75,000}{50,000} = \boxed{1.5 : 1}\end{aligned}$$

Meaning:

The firm has ₹1.50 current assets for every ₹1 current liability.

Liquid Ratio (Quick Ratio)

$$\text{Liquid Assets} = \text{Current Assets} - \text{Stock}$$

$$= 75,000 - 25,000 = ₹50,000$$

$$\text{Liquid Ratio} = \frac{50,000}{50,000} = \boxed{1 : 1}$$

QUESTION 4

From the following data, calculate:-

1. Debtors Turnover Ratio (Collection Period)
2. Creditors Turnover Ratio (Payment Period)

Information given:

- Debtors = ₹36,500
- Bills Receivable = ₹3,500
- Credit Sales = ₹7,30,000
- Creditors = ₹18,250
- Bills Payable = ₹1,750
- Credit Purchases = ₹3,65,000

Debtors Ratio (Collection Period)

$$\begin{aligned}\text{Debtors Ratio} &= \frac{\text{Debtors} + \text{Bills Receivable} \times 365}{\text{Credit Sales}} \\ &= \frac{(36,500 + 3,500) \times 365}{7,30,000} \\ &= \frac{40,000 \times 365}{7,30,000} = \boxed{20 \text{ days}}\end{aligned}$$

Creditors Ratio (Payment Period)

$$\begin{aligned}\text{Creditors Ratio} &= \frac{(\text{Creditors} + \text{Bills Payable}) \times 365}{\text{Credit Purchases}} \\ &= \frac{(18,250 + 1,750) \times 365}{3,65,000} \\ &= \frac{20,000 \times 365}{3,65,000} = \boxed{20 \text{ days}}\end{aligned}$$

QUESTION 5

debentures 100000,

bank loan for the five years 40000,
then the long term debts are 1,40000
If owners fund are 2,50000
so find out **debt equity ratio**.

$$\begin{aligned}\text{Debt Equity Ratio} &= \frac{\text{Long term liabilities}}{\text{Share holders fund}} * 100 \\ &= \frac{1,40000}{2,50000} * 100 \\ &= 0.56 \%\end{aligned}$$

For every rs. 100 share holders fund there is long term debts
of rs. 56.